

SPARC comparison II: low-surface-brightness and benchmark disks

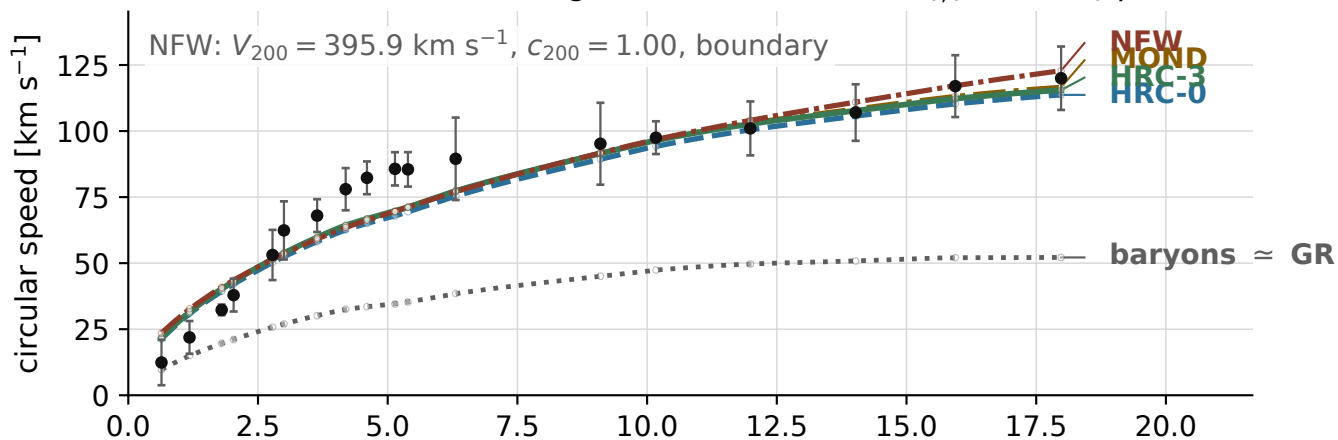
Black circles: SPARC. Smooth traces: residual-gated display curves; faint nodes: frozen model ordinates.

Fits and diagnostics use the unsmoothed nodes. HRC-0 $\equiv$ MOND-standard at equal scale.

Fixed scales: $a_M = 1.0824 \times 10^{-10}$ and $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$; NFW fitted per galaxy.

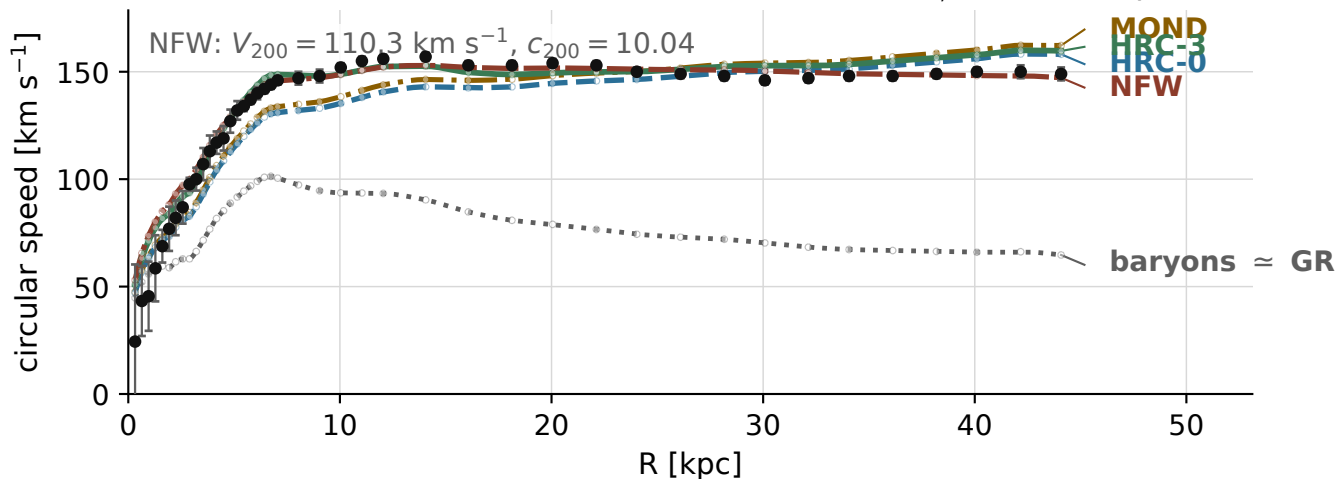
F568-3 — low-surface-brightness disk ($T = 7$; $\Sigma_{0,d} = 132 L_\odot \text{ pc}^{-2}$)

NFW: $V_{200} = 395.9 \text{ km s}^{-1}$, $c_{200} = 1.00$, boundary



NGC3198 — extended benchmark disk ($T = 5$; $\Sigma_{0,d} = 1603 L_\odot \text{ pc}^{-2}$)

NFW: $V_{200} = 110.3 \text{ km s}^{-1}$, $c_{200} = 10.04$



Λ CDM-motivated free NFW fit; not a unique Λ CDM prediction.

All panels: $Y_d = 0.5$, $Y_b = 0.7$, signed gas and a 2 km s^{-1} error floor.